

Supplementary Grading Rubrics for Open-Ended Project

Multiplayer Game Track (Cap A) — Online Trading Platform Track (Cap A+)

Tang Liang
DigiPen Institute of Technology

Purpose and Scope

This document is a **supplementary rubric** to clarify what different **letter grades** typically look like for the open-ended project options.

Grade caps by track.

- **Multiplayer Game Track (Option 1/2):** grade is capped at **A**.
- **Online Trading Platform/App (Option 3):** grade can reach **A+**.

Important notes (applies to both tracks).

- “More features” does not automatically mean a higher grade. **Correctness, robustness, and completeness of the end-to-end experience** matter more.
- Networking is a core requirement: handle concurrency, partial reads (TCP stream), loss/duplication/out-of-order (UDP), and clean error handling.
- Your implementation should be demonstrable with clear run instructions and predictable behavior.

Track 1: Multiplayer Game (Cap A)

A-level Feature Expectations (guidance list)

A project aiming for **A** should implement **most** of the following (well):

- Lobby/session flow (create/join room, player list, ready/start)
- Persistent user-specific data (money/XP/inventory/unlocks) across logins
- Asset selection/purchase/unlock with persistence
- Shared scoring / session leaderboard (global leaderboard optional)
- At least **3 players simultaneously** with meaningful interaction
- Clear game ending flow (message/results/animation/transition)
- BGM/sound effects (basic integration; volume control is a plus)
- In-game comms: text chat (good); audio chat (excellent, optional)

Letter-grade rubric (Game Track)

Track 2: Online Trading Platform/App (Cap A+)

Core functional expectations (baseline)

A viable trading platform/app should clearly support:

- User identity (at least username-based) and session handling
- Account state: cash balance and holdings (assets)
- Order placement (buy/sell) with validation (symbol, qty, price)
- Matching/pricing logic (choose one model and document it):
 - simplified order book (price/time priority), or
 - market-maker quotes (bid/ask), or
 - periodic auction/clearing
- Consistent server-authoritative updates (no double execution, no negative balances unless explicitly allowed)

Grade	Typical characteristics (Game Track)
A	Complete, polished multiplayer experience. Robust multi-client play (3+), consistent shared scoring, clean session flow (lobby or equivalent), and at least one meaningful persistence feature (e.g., inventory/XP/currency). Network behavior is stable (no obvious desync/jitter breaking gameplay). Disconnects handled reasonably (graceful leave; no crash/hang). Clear end-of-game state. Sound/UI integration does not compromise correctness.
A-	Strong and mostly complete. Core multiplayer gameplay works well and is stable, but one area is weaker: e.g., persistence is minimal or slightly unreliable; lobby is basic; end-of-game flow is functional but not polished; occasional jitter/desync but still playable; limited resilience on disconnect/rejoin.
B+	Solid core, partial completeness. Multiplayer gameplay works for 3+ but with noticeable rough edges: limited session flow, incomplete persistence, inconsistent leaderboard/scoring updates, or minor protocol/design issues. Generally works in standard demos, but robustness under edge cases (disconnects, rapid actions) is not strong.
B	Playable but missing several A-level elements. Basic multiplayer connection and interaction works, but limited features and/or shaky robustness. May rely heavily on client trust (weak authoritative scoring), or show frequent sync artifacts. Limited polish in UX flow.
B-	Minimum multiplayer satisfied but fragile. Demonstrates multiplayer concept, but frequent issues: instability, inconsistent state across clients, unclear game lifecycle, weak error handling, and incomplete testing.
C+	Partially working. Some multiplayer functionality exists, but core requirements are incomplete or unreliable. Limited demonstration success or requires significant manual intervention to run.
C	Major gaps. Basic networking/game logic incomplete; frequent failures; cannot reliably demonstrate the required multiplayer experience.
D+	Very limited. Small portion of functionality works (e.g., connection only), but gameplay/networking largely incomplete.
D	Barely functioning. Minimal progress; cannot demonstrate meaningful multiplayer gameplay.
F	Fail. Not submitted, not runnable, or does not demonstrate the required networking functionality.

Table 1: Letter-grade rubric for the Multiplayer Game track (grade cap A).

- A usable UI: market view + account view + notifications/errors

Letter-grade rubric (Trading Track)

Quick checklist (what improves grades for both tracks)

- Clear architecture separation: networking vs logic vs UI vs persistence
- Robust parsing/protocol handling (TCP stream partial reads; UDP loss/dup/out-of-order)
- Server-authoritative state for key outcomes (score/trades/balances)
- Graceful handling of disconnects and errors (no crash/hang)
- Reproducible demo package: README + configuration + test steps

Grade	Typical characteristics (Trading Track)
A+	Excellent, production-like robustness. Correct trading logic with clear matching/pricing model, strong consistency (atomic updates to balances/holdings/orders), clean handling of concurrency and edge cases (rapid orders, partial fills if supported, cancellations). Good resilience: reconnect restores state (snapshot refresh at minimum). Clear audit trail (trade/order history) and strong documentation (protocol format + run steps + test checklist).
A	High quality and correct. Core trading flows work reliably: place orders, execute trades, update accounts correctly, and show consistent state across clients. Good validation and reasonable robustness, though one advanced aspect is weaker (e.g., limited reconnect handling, limited history/audit, simplified matching rules, or less polished UI).
A-	Mostly correct with minor weaknesses. Trading system works in normal scenarios, but has notable limitations: incomplete order states, weaker handling of concurrency, occasional state inconsistency under stress, or limited clarity in UI/UX. Documentation adequate but not strong.
B+	Functional core but limited robustness. Basic trading works (submit buy/sell and update balances), but matching/pricing is simplistic and/or reliability is limited. Edge cases may break (duplicate requests, rapid submissions, disconnect). UI is basic.
B	Works in demos but incomplete. Demonstrates a trading concept, but with missing features or correctness gaps (e.g., weak validation, inconsistent holdings, unclear matching). Requires careful demo conditions.
B-	Fragile. Some trading functions exist, but correctness/reliability is weak; inconsistent state appears; limited ability to handle multiple users or concurrent actions.
C+	Partially working. One or more major functional pieces are incomplete (e.g., matching, account consistency, multi-client). Demo is unreliable.
C	Major gaps. Cannot demonstrate a coherent end-to-end trading workflow with correct state updates.
D+	Very limited. Minimal UI or network communication exists, but trading logic/state management largely missing.
D	Barely functioning. Minimal progress; no meaningful trading workflow.
F	Fail. Not submitted, not runnable, or does not demonstrate the required networking functionality.

Table 2: Letter-grade rubric for the Online Trading Platform/App track (can reach A+).